

SYSTEMS DESIGN

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

1 Requirements Engineering

2 Design & Process



Outline

1 Requirements Engineering

2 Design & Process



Stakeholders Vs. Shareholders

- **Stakeholders** are individuals or groups who have an interest in the success of a project.
- **Stakeholders** can be internal or external to a company. For example, customers, employees, suppliers, and regulators are external stakeholders.

Stakeholders are individuals or groups who have an ownership interest in a company.

Shareholders are internal to a company. For example, investors, owners, managers are internal stakeholders.

devs
legal
sales
sponsors
final users



Stakeholders Vs. Shareholders

- **Stakeholders** are individuals or groups who have an interest in the success of a project.
- **Stakeholders** can be internal or external to a company. For example, customers, employees, suppliers, and regulators are external stakeholders.
- **Shareholders** are individuals or groups who have an ownership interest in a company. *av.*
- **Shareholders** are internal to a company. For example, investors, owners, and managers are internal stakeholders.



Requirements

clear

- **Requirements** are (statements) that describe the features, functions, and constraints of a system.
- **Requirements** are used to communicate the needs of stakeholders to developers.
- Requirements are used to guide the design, development, and testing of a system.

client

needs
(problem)

Devs.

(software
solution)

design

requirement



Requirements

- **Requirements** are statements that describe the features, functions, and constraints of a system.
- **Requirements** are used to communicate the needs of stakeholders to developers.
- **Requirements** are used to guide the design, development, and testing of a system.

T.D.P.

Test-driven Development



User Stories → not-technical

- **User stories** are short, simple descriptions of a feature or function of a system.
- They are written from the perspective of the user and describe what the user wants to achieve.
- They are used to capture the requirements of a system in a simple and understandable way.



User Stories: Format Example

T.D.D.

Product Owner

User Story

+ Pre-requisite

no time effort

Impact

Title: function	Priority:	Estimate:
User Story: <u>As a</u> [description of user], <u>I want</u> [functionality] <u>so that</u> [benefit].		
Acceptance Criteria: Given [how things begin] When [action taken] Then [outcome of taking action]		

Test

As a student → **role**

I want enroll a course **req.**

so that I can finish my course.

scenario test

expected result

Unit

int

stress

system

ProductPlan



What is Requirements Engineering?

step

- **Requirements engineering** is the process of eliciting, analyzing, specifying, validating, and managing the requirements of a system.
- It is a critical activity in the systems development lifecycle that ensures that the system meets the needs of its users.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

solve → problem



What is Requirements Engineering?

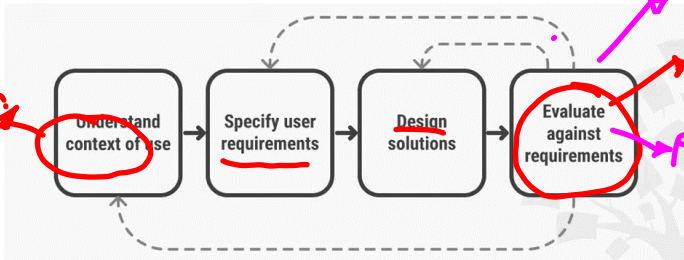
- **Requirements engineering** is the process of eliciting, analyzing, specifying, validating, and managing the requirements of a system.
- It is a critical activity in the systems development lifecycle that ensures that the system meets the needs of its users.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

expectations
align



User-Centered Design (UCD)

- **User-centered design** (UCD) is an iterative design process that focuses on understanding the needs, preferences, and behaviors of users.
- UCD is a collaborative process that involves users in the design and development of a system.
- UCD is used to create systems that are usable, efficient, and satisfying to users.



Requirements Engineering Process

The **requirements engineering** process consists of the following activities:

- Gathering requirements.
- Analyzing requirements.
- Validating requirements.
- Verifying requirements.
- Documenting requirements.
- Managing requirements.
- Communicating requirements.

client comm.
system

mockups
diagram

software test
user stories + UML

User story
mapping UJM



Gathering Requirements

- **Gathering** requirements is the process of **collecting** and **documenting** the **needs** of stakeholders.
- It involves interviewing stakeholders, conducting surveys, and observing users to understand their requirements.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

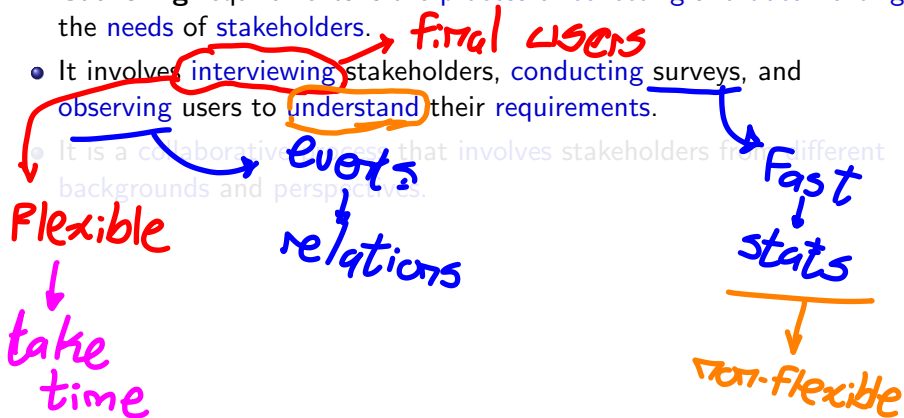
problem(s)

↓
root



Gathering Requirements

- **Gathering** requirements is the process of collecting and documenting the needs of stakeholders.
- It involves interviewing stakeholders, conducting surveys, and observing users to understand their requirements.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.



Gathering Requirements

- **Gathering** requirements is the process of collecting and documenting the needs of stakeholders.
- It involves interviewing stakeholders, conducting surveys, and observing users to understand their requirements.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

expectations
align

negotiation



Clients are not always right

Dear Santa,
 How are you? I'm good.
 Here is what I want for
 Christmas.

A http://www.amazon.com/gp/product/B0032HFG0M/ref=59_hps_bw_g21-ir03?pf_rd_m=ATVPDKIKXODER&pf_rd_s=center-3&pf_rd_f=1XW442FH2K03Y7BMWQNM&pf_rd_t=101&pf_rd_p=1328901542&pf_rd_i=165379

//
 C



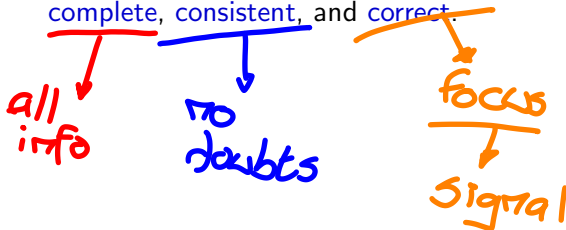
Analyzing Requirements

- **Analyzing** requirements is the process of examining and understanding the requirements of a system.
- It involves identifying dependencies, conflicts, and inconsistencies in the requirements.
- It is a critical activity that ensures that the requirements are complete, consistent, and correct.



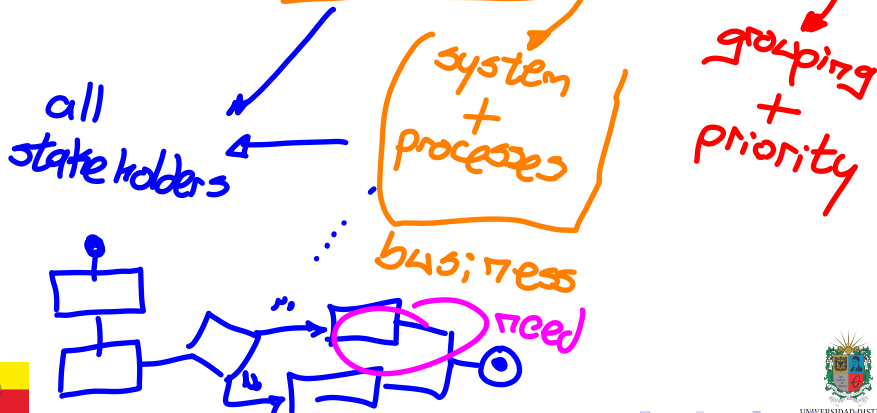
Analyzing Requirements

- **Analyzing** requirements is the process of examining and understanding the requirements of a system.
- It involves identifying dependencies, conflicts, and inconsistencies in the requirements.
- It is a critical activity that ensures that the requirements are complete, consistent, and correct.



Documenting Requirements

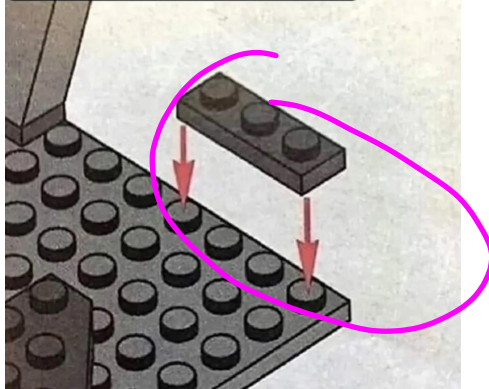
- **Documenting** requirements is the process of writing and organizing the requirements of a system.
- It involves creating documents, diagrams, and models that describe the requirements in a clear and concise way.



Everyone hates to write Documentation

**En la documentación
está todo bien explicado**

La documentación:



Validating Requirements

- **Validating** requirements is the process of ensuring that the requirements are correct and complete. *client*
- It involves reviewing the requirements with stakeholders to verify that they meet their needs.

client
↓
satisfaction

tech team
↓
feasible



NOT Clear Understanding of Requirements



Dad Jokes
@Dadsaysjokes

My dad told me his password is:
MickeyMinnieGoofyDonaldPlutoHuey
LouieDeweyDublin.

Because he was told his password
had to contain 8 characters and at
least one Capital.

NO
Understanding

example
training



Verifying Requirements

- **Verifying** requirements is the process of ensuring that the requirements are correctly implemented in the system.
- It involves testing the system to verify that it meets the requirements.
- It is a critical activity that ensures that the system meets the needs of its users.

product $\Rightarrow$ testing
engineering

- check functionalities
- quality



Verifying Requirements

- **Verifying** requirements is the process of ensuring that the requirements are correctly implemented in the system.
- It involves testing the system to verify that it meets the requirements.
- It is a critical activity that ensures that the system meets the needs of its users.

unit
integration
stress
system
acceptance

solve
the problem

QA - area
testing
org. area / rol



Typical Mistakes when Testing

Disturbing Chinese calorie app...



	カシューナッツ (cashew)	1粒	9 kcal
	ジャムパン (Pomeranian)	1個	327 kcal

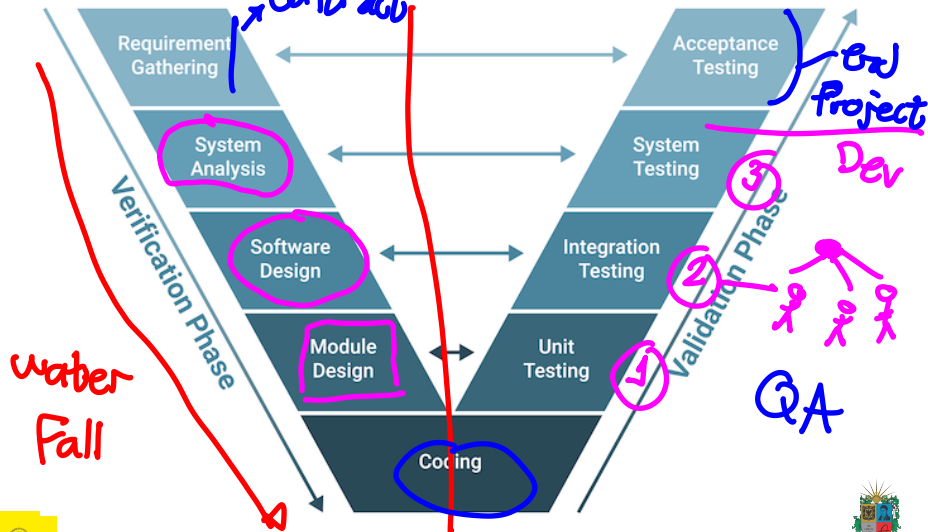


V-Model in SDLC

70s-80s

to Use

Contract



Outline

1 Requirements Engineering

2 Design & Process



Conceptual Design

- Once the initial set of requirements is defined, the next step is to create a conceptual design of the system.
- Conceptual Design** is a high-level design that defines the structure and behavior of the system. It is achieved by the recognition of the appropriate components, connections, and responsibilities.
- The conceptual design is used to communicate the vision of the system to stakeholders and to guide the development of the system.

tech-less

understand

Problem as a system



Conceptual Design

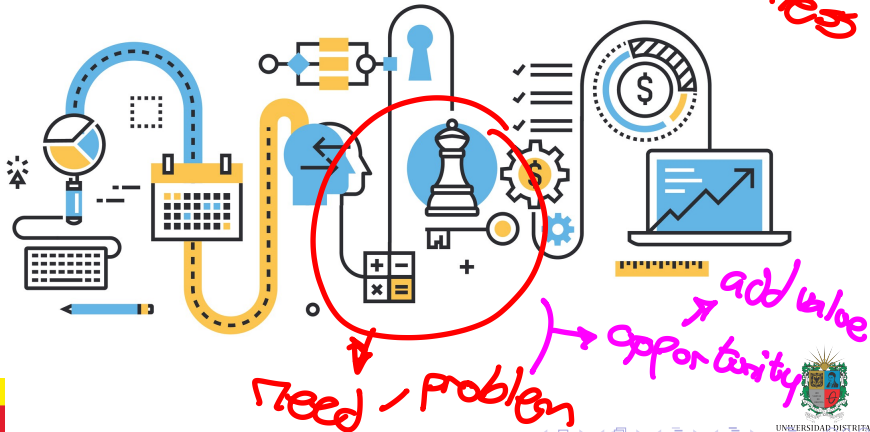
- Once the initial set of **requirements** is defined, the next step is to create a **conceptual design** of the system.
- **Conceptual Design** is a **high-level design** that defines the structure and behavior of the system. It is achieved by the recognition of the appropriate **components**, **connections**, and **responsibilities**.
- The **conceptual design** is used to **communicate** the **vision** of the system to **stakeholders** and to **guide** the **development** of the system.

client
Diagram



Process Definition

- A **Process** is a series of steps or actions taken to achieve a particular end.
- **Processes** are used to organize and manage work.



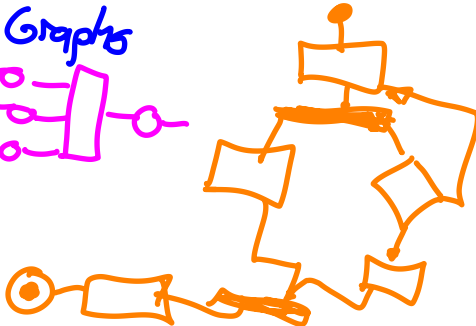
Workflows → Pipelines → automation

- A **Workflow** is a series of tasks that are performed in a specific order to achieve a goal.
- **Workflows** are used to automate and optimize business processes.
- **Workflows** can be sequential, parallel, conditional, or repetitive.

↳ process-based



Graphs



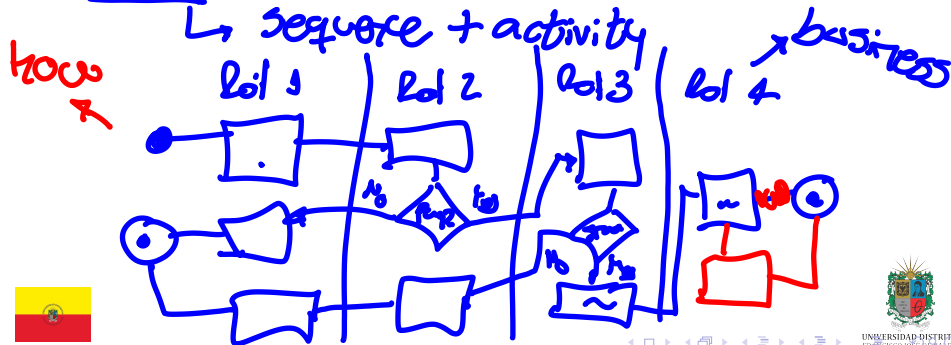
② quality
① speed

Flowchart

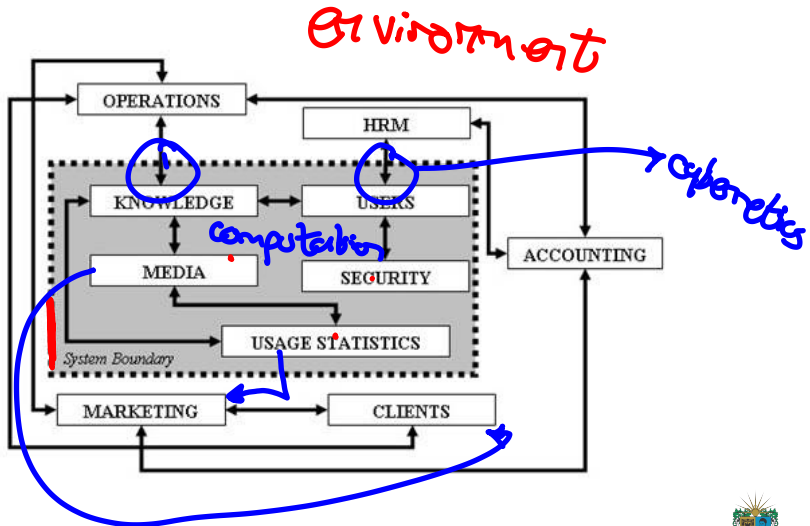


Process Models

- A Process Model is a representation of a process that shows the sequence of steps and the relationships between them.
- Process models are used to analyze, design, and improve processes.
- Examples of process models include flowcharts, data flow diagrams, activity diagrams, business process model and notation (BPMN), petri nets, state diagrams, among others.

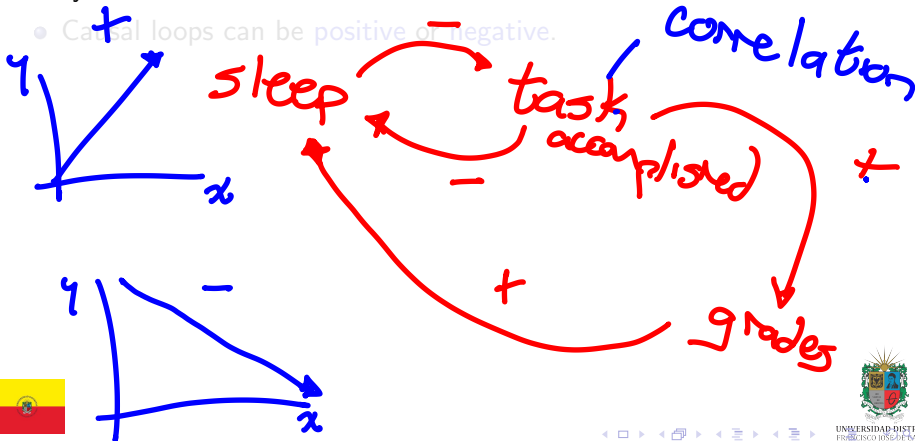


System Schema Example: Company Structure



Causal Loops

- A **Causal Loop** is a **diagram** that shows the **relationships** between different variables in a system.
- Causal loops are used to **analyze** and **understand** the **dynamics** of a system.
- Causal loops can be positive or negative.



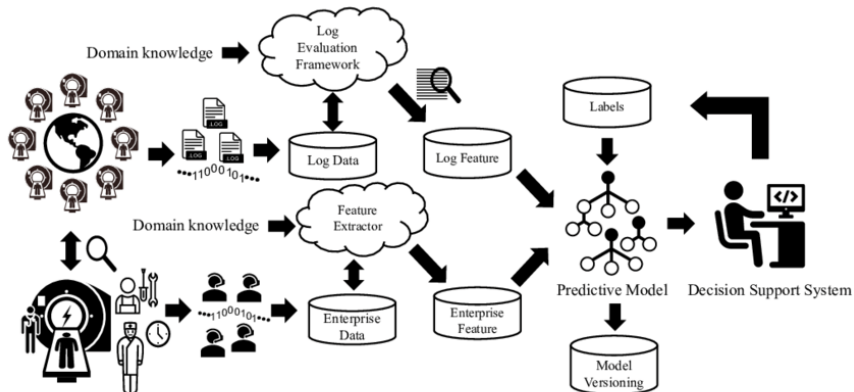
Causal Loops

- A **Causal Loop** is a **diagram** that shows the **relationships** between different variables in a system.
- Causal loops are used to **analyze** and **understand** the **dynamics** of a system.
- Causal loops can be **positive** or **negative**.

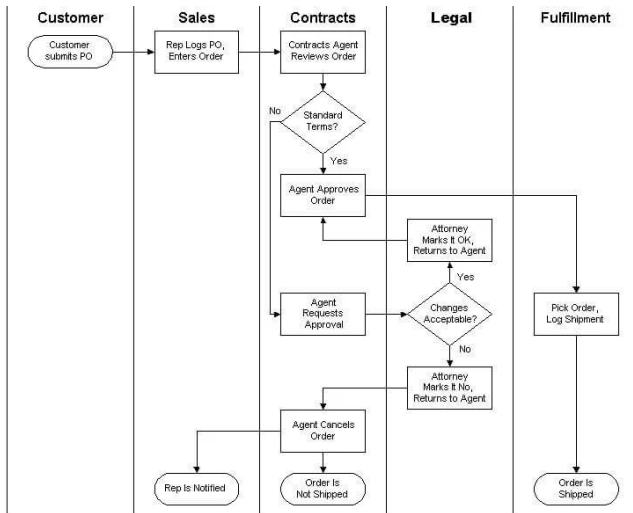
balance → stability
↓
chaotic pattern



System Schema Example: Processing Pipeline



Business Process Model and Notation (BPMN)



Technical Design

- Once the **conceptual design** of the system is defined, the next step is to create a **technical design** of the system.
- **Technical Design** is a **detailed design** that defines the architecture, components, and interfaces of the system.
- The **technical design** is used to **guide the development** of the system and to **communicate the implementation details** to **developers**.

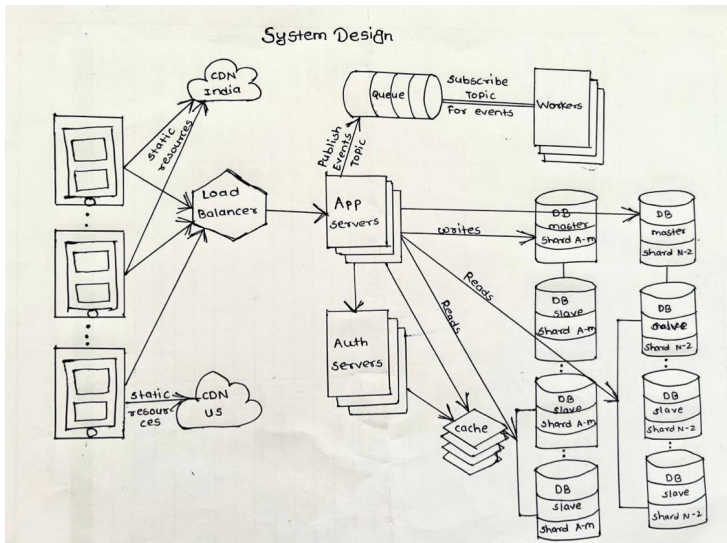


Technical Design

- Once the **conceptual design** of the system is defined, the next step is to create a **technical design** of the system.
- **Technical Design** is a **detailed design** that defines the architecture, components, and interfaces of the system.
- The **technical design** is used to **guide** the **development** of the system and to **communicate** the **implementation** details to **developers**.



Systems Design applied to Software Architectures



Outline

1 Requirements Engineering

2 Design & Process



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

